

INTRODUCTION TO ASTRONOMY

Mon and Wed | 10:00a-11:15a | MEZ 2.104 or BUR 124

Instructors

Prof. Steven Finkelstein

(he/him/his)

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Help Session (HS):

M 1:00-2:00p

PMA 16.316 & Zoom:

<https://utexas.zoom.us/j/94113222092>

Assistant Instructor:

Laurel Weiss

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In-person review sessions:

M & W 5-6p @ PMA 15.216B

HS: T & Th 1-2p @ PMA 16.318

Teaching Assistants

TA: Joohyun Lee

joohyun.lee@utexas.edu

HS: T/Th 11a-12p, PMA 16.206

Learning Assistants

TBD

TBD@utexas.edu

HS: TBD

What is intro to astronomy about?

This introductory course for non-science majors will survey all of astronomy, starting with the ancient Greeks and how their studies led to the correct model for the Solar System, moving on to studying other stars and planets. We will finish with learning about our own Milky Way galaxy, and peering back into the early phases of the Universe to see early galaxies and understand the Big Bang. By participating in this class, ***you will develop*** an appreciation for astronomy and the night sky around you. ***You will improve skills*** in critical thinking, communication and teamwork.

This course carries the ***Quantitative (QR) Reasoning flag***. QR courses are designed to equip you with skills that are necessary for understanding the types of quantitative arguments you will regularly encounter in your adult and professional life. You should therefore expect a substantial portion of your grade to come from your use of quantitative skills to analyze real-world problems.

Course Learning Objectives: By fully engaging in this course, you will...

Develop a broad understanding of the nature, scope and evolution of the Universe, and where the Earth and Solar System fit in.

Improve your critical thinking and quantitative reasoning skills, and their importance in the context of the scientific process

Learn that science is a process, the world is knowable, and we come to know it through observations, experiments and theory.

Get acquainted with the history of astronomy and the evolution of scientific ideas (science as a cultural process).

Gain a familiarity with the night sky and how its appearance changes with time and position on Earth.

Each class period will have learning objectives linked to one or more of these course goals

Email Policy: For questions about class content or logistics, please first look to this syllabus where we address common questions. If you don't find the answers there, you may email: Joohyun for technical support questions, and ULA TBD for Canvas grading questions. They'll do their best to get back to you within one business day. For questions of a personal nature please email AI Laurel; she will loop Prof Finkelstein in if necessary.

Class Format:

- You are in an in-person section of a class that is simultaneously live-streamed to a web-based section. You will be pre-assigned to either the Mezes or Burdine classroom, which you will attend throughout the semester. Your responsibilities will depend on which room you are in, and will differ in places from the web-based class, as detailed below. Some specifics:
 - Both classrooms are active. This means during activity times you are actively engaged with your groups.
 - You will use ABCD voting cards (available on Canvas) for in-class voting and not instapoll.
 - I expect you to ask questions frequently.
 - **Mezes only:** When I ask questions, I expect lots of volunteers! If I find you consistently not volunteering, you risk receiving a zero for engagement, and I will ask you to switch to the web-based section.
 - **Mezes only:** You will be required to do one in-person, live on-camera astronomy in the news presentation (info below).
 - **Burdine only:** You will be chat Group Leaders, responsible for communicating questions from your chat, and holding study sessions before each exam.

Class recordings: These are reserved only for the use of members of this class (students, TAs, and the instructor) and only for educational purposes. Recordings should not be shared outside the class in any form.

What is expected of me in this class?

- This is an active-learning classroom, meaning that you must attend class and **participate!** Work collaboratively and be prepared to share your ideas. If attending class during scheduled class time is a problem, please find another course which fits your schedule.
- Complete all assigned modules and assignments on time. Make sure to take time to think deeply about the videos, and spend time with the book or come get help if you have questions before starting the homework!
- Start the projects early! Everything you need to know is in this syllabus or on the Canvas assignment page, so you're welcome to get them done early!
- Take advantage of your teaching team, we are here to help you!

What happens in lecture?

- Not much lecturing! I have designed this class from the ground up to be very interactive, focusing on your attainment of the course learning outcomes. You will be exposed to the content outside of class through Canvas homework modules, which replace the typical lecture component.
- In class is time to work! We will review concepts you've learned prior to coming to class through Canvas homework modules, and you will work with your peers in class to practice these concepts with engaging activities.
- You will only learn if you participate, thus attendance and participation are **required!**
- A typical class day will be composed of the following:
 - Astronomy in the news.
 - Instapoll questions to review the module
 - Several discussion questions.
 - Answering questions from you about the topics which may still be unclear.
 - Activities in groups, followed by whole class discussion.

My Commitment to all Students Succeeding

The University of Texas at Austin serves a diverse student body, and it is my intent that all students are able to succeed in this course, regardless of their background. We are different in many ways, and this is fundamental to building and maintaining an inclusive campus community. Astronomy belongs to all people, independent of race, religion, gender, gender identity, gender expression, or sexual orientation. Incidents of discrimination, assault, harassment, threats, intimidation, profiling, or coercion based on membership or perceived membership will not be tolerated. Show each other respect no matter perceived knowledge or performance in this class, or any other.

I pledge to treat all of you with respect, and I ask that you do the same in return to myself and to your classmates. In the classroom, I will gladly honor your request to address you by a name that is different from what appears on the official roster, and by the gender pronouns you use. I ask you to support one another as you talk about differences in your personal experiences whether relevant to the class or not.

Study Groups

We are piloting a new concept for this classes - the bulk of students will be online, but 50 students who signed up for the in-person section will meet in-person live during the class (20 in the Mezes studio with Prof Finkelstein, and 30 in the Burdine studio with a TA). *All* students (in-person and online) will belong to a study group. The Group Leader will be a student in the Burdine classroom. Each online student will be placed into a small-group chat with the Group Leader, where you will work on the in-class activities. During the class the leader's job will be to help communicate questions from their chat to the class during discussion time. They will also organize study sessions (in-person, on zoom, or hybrid) prior to each exam (more below under participation grading). The aim of these study groups is to help build connection, and enhance both your understanding of the material and your enjoyment of the course. Each group will be in competition throughout the semester to earn the most "group points", to earn a special prize of "astronomy swag" at the end of the semester. Each group will earn one point for every time a group member attends a help session and asks a class content-related question; more point opportunities may become available throughout the semester.

How is my performance in this class assessed?

What is the grading scale?*

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93.0 - 100 A
90.0 - 92.99 A-
87.0 - 89.99 B+
83.0 - 86.99 B
80.0 - 82.99 B-
77.0 - 79.99 C+
73.0 - 76.99 C
70.0 - 72.99 C-
67.0 - 69.99 D+
63.0 - 66.99 D
60.0 - 62.99 D-
< 59.9 F

Your final course grade will be determined as follows:

35% - Engagement:

This has several components:

In-class participation (20%): You will receive credit for this attending class, and actively engaging in discussions and answering in-class "think-pair-share" questions. Be sure to sign in with your classroom TA when you arrive.

• **Astronomy in the News (Mezes Students; 10%):** You will research a recent news item in the field of astronomy which is interesting to you, and prepare a news-style video presentation, which you will presents live, in class. You must sign up during the first week of class on the in-class news calendar to reserve your date! At least two days prior to your presentation, email both Prof Finkelstein and Laurel, including the date you wish to present, your article title, the link to your article, attach the image you would like shown. Full details are available on the canvas assignment.

- **Study Groups (Burdine Students, 10%):** You will serve as a chat group leader. In addition to leading the discussions in your chat, and bringing questions to the attention of Prof Finkelstein, you will organize study sessions prior to each exam. You are encouraged to, for each session, poll your group members for an ideal time - it may not be possible to meet everyone's schedule, but try your best (those who cannot make it can attend another group's session). You will receive credit for this component by submitting either a selfie-photo of you with your group (either in person, or via zoom), with obvious evidence that astronomy is being studied.

Although ***makeup participation points will not allowed***, I realize that you may need to occasionally miss class. For this reason, Canvas will automatically ***drop your three lowest attendance/engagement grades***. This large number of drops is to account for unexpected absences, including illness. ***Should you miss a few classes due to illness (or any other reason), there is no need to contact us.*** Your zero grade on those days will be dropped, and you can catch up on what you missed with the recordings on Canvas. Should illness or other circumstances cause you to miss two or more class periods in a row, please contact Student Outreach and Support, and they will contact Laurel and Prof Finkelstein for accommodations.

15% - Projects: There are two projects, each worth 7.5% of your total grade. Full details for each assignment are listed on the Canvas assignment pages, but a listing of the projects is provided here with a due date (any can be turned in early!):

- Project #1 - Online students only, please ignore!
- Project #2: **Moon Journal (due April 1)**: Observe the Moon over the course of one phase cycle, and keep track of the phases in a journal. This will be turned in digitally, though a physical in-person submission is possible.
- Project #3: **Citizen Science (due Apr 15)**: You will participate in an online citizen science activity, and turn in a reflection.

25% - Homework (10% pre-class; 15% assessment): Each class will have a Canvas module, with reading and videos assigned to be completed prior to each class. Each module will include a Pre-class Homework, done via Canvas quizzes, to assess your retention of the information presented in the module reading. Please take detailed notes!! You may take the Pre-class Homeworks up to two times, and your final grade will be the average of your two attempts. To deter cheating, these canvas quizzes will make use of the Respondus lockdown browser, meaning that once you start the quiz you cannot click away from the browser window until you are complete.

At the end of most weeks we will have an assessment Homework to assess your understanding of the topic after we have worked on it in class. **These assessment Homeworks are designed to prepare you for the exam, so we urge you to study prior to taking them. You may only take them once.** You may use any notes you have taken. These will also use the Respondus lock-down browser software.

Your three lowest Pre-class Homework scores will be dropped, as well as your single lowest assessment homework score. If you miss four Pre-class Homeworks, then one will count as a zero. ***There will be no extensions.*** You are encouraged to study together in groups, but the quizzes should be done alone.

30% - Exams: We will have four closed-book exams, each covering roughly 1/4 of the course material. These will be held **in-person** and will consist of 20-30 multiple choice questions. Please make arrangements to be available on campus to take the exams during the times below! The exam times are:

Exam 1: February 10th 7:00-8:00pm, WEL 3.502

Exam 2: March 5th 7:00-8:00pm, WEL 3.502

Exam 3: April 2nd 7:00-8:00pm, WEL 3.502

Exam 4: April 27th 7:00-8:00pm, WEL 3.502

Each exam will have a pre-class review sessions in the evening prior to the exam; time and location TBD.

Make-up exams will not be offered, but your single lowest exam grade will be dropped. We will identify a single alternate exam time only for documented emergencies or official university travel. For those who qualify but cannot make this alternate time they may use their exam drop.

Extra Credit Project: In addition to the exam study session opportunity, there one final opportunity for extra credit: Astronomical Observing. You will need to visit the telescope on the roof of Painter Hall during one of the Friday and Saturday night public viewing events, or the roof of PMA during one of the Wednesday night events, and **submit a video recording of yourself on-site**, discussing what you observed. Information on the observing nights can be found here: <http://outreach.as.utexas.edu/public/viewing.html>. This is due no later than **Apr 27th**. The maximum grade for this component is 5 points added to your exam score total. Please see the Canvas assignment for full details.

What are other policies on exams, assignments, and other course structure?

Course Website:

Canvas page for this course:

TBD

Course Webpage: The course webpage on the Canvas system will be updated with course announcements, homework and reading assignments, and deadlines. It is your responsibility to check these on a regular basis. Please come to class prepared, having done the assigned module. Also please be prepared to participate in in-class discussions and activities, this is for your benefit.

Late work: Late work is not accepted. Makeup (or early) homework or exams are not offered. *However*, I understand that life events happen, so if you are unable to turn an item in on time or attend a Homework or exam, contact the **Laurel in advance of the due date**. Note that being busy with other classes or work will not be considered a valid excuse. If you miss class for a sponsored University event, and you contact Laurel **in advance of the due date**, we can discuss accommodations.

Course Conduct: Please minimize distractions so you can stay focused. During the activity time, please be active with your chat group, and make the best use of your time to get through the material. **Be respectful of others**, especially during in-class peer discussion times, and even if you disagree with them. No late arrivals or early departures are allowed. Please silence cell phones before you enter the classroom.

Extra credit: Other than the options discussed above, there will be no opportunities for extra credit. Please don't email us at the end of the semester asking for additional points (this includes grade rounding!).

Where can I find.. ?

Canvas will have:

1. Important announcements
2. Access to the Video Stream
3. Lecture slides
4. Assignments & modules
5. Syllabus
6. List of Learning Objectives
7. Gradebook

Canvas will be our main form of communication, so check it regularly and stay up to date on assignments and communications.

Students with Children: I recognize the difficulty of being a full time student with children. If you have children, or other family commitments, please come see me to discuss any modifications of the course policies which will maximize your success in this course.

Email: Email is recognized as an official mode of university correspondence; therefore you are responsible for reading your email for university and course-related information and announcements. Please check your email regularly and frequently (this includes Canvas announcements).

Use of AI:

The Use of AI: The creation of artificial intelligence tools for widespread use is an exciting innovation. These tools have both appropriate and inappropriate uses in classwork. The use of artificial intelligence tools (such as ChatGPT) in this class shall be permitted on a limited basis. Specifically:

- You may use Chat GPT (or any tool) to help find information and references while studying. Be warned it is often inaccurate.
- You may **not** use Chat GPT (or any other tool) for writing - all words should be your own.
- You may **not** use Chat GPT (or any other tool) during Canvas quizzes (and of course, not during the in-person exams).
- Important: **Any** time AI tools are used, they should be used with caution and proper citation, as the use of AI should be properly attributed. Using AI tools without my permission or authorization, or failing to properly cite AI even where permitted, shall constitute a violation of UT Austin's Institutional Rules on academic integrity.

If you are considering the use of AI tools but are unsure if you are allowed or the extent to which they may be utilized appropriately, please ask.

Illness-related absences

We understand that illness often results in absences. This course has significant built-in flexibility, allowing you to miss up to five classes before your participation grade is impacted. **Should your illness extend more than a week**, please reach out to Student Outreach and Support to provide illness documentation. Please do not share your medical information with me or the TAs. Since course requirements and flexibility will vary between classes, schools and colleges, you are responsible for understanding expectations of your faculty members and any accommodations offered.

Frequently Asked Questions:

Do you record the in-class lectures?

Yes! These recordings are useful to review the material, the video of our instapoll discussions, or the debrief after the activities. The video stream will be available after class via the Lectures Online page Canvas.

How do I succeed in this class?

The best way to succeed is to prepare and participate. Do the modules ahead of time, and take the time to watch them in full (and multiple times if needed). When in class, buy in and ***participate!***

How do I study for the exams?

- 1) Study the activities. Don't just read them, re-do them! Work in a group if you can!
- 2) Go over the in-class PDFs, and practice the instapoll questions.
- 3) Re-watch the relevant module videos, find concepts which you feel less secure on, find those concepts in the book, and read up.
- 4) Come to help sessions!!!

I missed an exam, when can I make it up?

There are no makeup exams. The only exception to this will be:

- 1) If you have a major life event, **and you notify Laurel ahead of time.** Depending on the situation, we may ask you to contact Student Emergency Services for assistance.
- 2) I am contacted by Student Outreach and Support, and they authorize a makeup.
- 3) You are absent for a university-sponsored event, ***and*** you notify Laurel ahead of time. In these cases, you are responsible for being available during the alternate time slot.

I'm sick, and can't attend class today, what do I do?

Don't email us! Rest and get better (you can miss up to five classes, and still receive a full participation grade). Do the module before class, and after class watch the video to see what you missed. Do the missed activities, in a group if you can find some classmates, or on your own. If you have to miss multiple consecutive classes, please contact Student Outreach and Support, and they will let us know if they feel you should be allowed to make up for missed participation.

I need to leave class early? How do I make sure I don't lose participation?

Participation is counted through InstaPoll and the chat. You will not receive credit for questions when you are not in class.

I forgot to turn in my project! Can I turn it in late?

No, late projects are not accepted. Don't let this be you, turn it in early!

Its two days before the moon journal is due, and I haven't started!

There's nothing I can do to help you. Don't let this be you - do this early in the semester!

I got a zero in the gradebook for something I did or turned in!

We can make mistakes when inputting hundreds of grades! If you believe there is a mistake in the gradebook, stay calm, just send the designated Canvas TA an email, and we'll investigate.

University Resources:

Academic accommodations (D&A): This class respects and welcomes students of all backgrounds, identities, and abilities. If there are circumstances that make our learning environment and activities difficult, or if you have medical information that you need to share with me, please let me know. I am committed to creating an effective learning environment for all students, but I can only do so if you discuss your needs with me as early as possible. I promise to maintain the confidentiality of these discussions. Any student with a documented disability who requires academic accommodations should contact Disability & Access at 512-471-6259 (voice) or 512-410-6644 (Video Phone) as soon as possible to request an official letter outlining authorized accommodations (email: access@austin.utexas.edu). For more information, visit <https://diversity.utexas.edu/disability/>. *I am also happy to meet to discuss more.*

Counseling and Mental Health Center: Do your best to maintain a healthy lifestyle this semester by eating well, exercising, getting enough sleep and taking some time to relax. This will help you achieve your goals and cope with stress. All of us benefit from support during times of struggle. You are not alone. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is often helpful. If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. <http://www.cmhc.utexas.edu/individualcounseling.html>

The Sanger Learning Center: Did you know that more than one-third of UT undergraduate students use the Sanger Learning Center each year to improve their academic performance? All students are welcome to take advantage of Sanger Center's classes and workshops, private learning specialist appointments, peer academic coaching, and tutoring for more than 70 courses in 15 different subject areas (including Astronomy). For more information, please visit <http://www.utexas.edu/ugs/slc> or call 512-471-3614 (JES A332).

University and Course Policies:

Academic integrity: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community. Students who violate University rules on scholastic dishonesty are subject to disciplinary penalties. Ethical conduct is expected at all times. For example, answering Voting Questions to receive credit when you are not in class is unethical. Incidences of academic dishonesty will be reported to Student Judicial Services. For more specific information go to: <http://deanofstudents.utexas.edu/conduct/academicintegrity.php>.

You are responsible for understanding UT's Academic Honesty and the University Honor Code which can be found at the following web address: <https://deanofstudents.utexas.edu/conduct/standardsconduct.php>

- **Sharing of Course Materials is Prohibited:** No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (homeworks, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.
- **Class Video Recordings:** Class recordings are reserved only for the use of members of this class (students, TAs, and the instructor) and only for educational purposes and are protected under FERPA. Recordings should not be shared outside the class in any form. Violation of this restriction could lead to Student Misconduct proceedings.

Plagiarism: As a research university, the University of Texas at Austin takes plagiarism very seriously. Do not risk getting involved in a plagiarism infraction - the consequences simply aren't worth it. Always cite your sources, and when in doubt consult a professor or librarian. You may also read more about plagiarism at the Student Judicial Services website: <http://deanofstudents.utexas.edu/conduct/academicintegrity.php>

Personal or Family Emergencies: If you experience a personal or family emergency (death in the family, protracted sickness, serious mental health issues) that prevents you from attending an exam or forces you to miss multiple days of class, you should contact Student Emergency Services in the Office of the Dean of Students <http://deanofstudents.utexas.edu/emergency/>. They will work with you to communicate with your professors and let them know of your situation.

Religious Days: A student who is absent from a class or examination for the observance of a religious holy day will be permitted to make up the missed work, if notice is given at least fourteen days prior to such an absence.

Core curriculum: This course may be used to fulfill three hours of the natural science and technology component of the university core curriculum and your successful participation addresses the following four core objectives established by the Texas Higher Education Coordinating Board: communication skills, critical thinking skills, teamwork, and empirical and quantitative skills.

Title IX: Beginning January 1, 2020, Texas [Senate Bill 212](#) requires all employees of Texas universities, including faculty, report any information to the [Title IX Office](#) regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, visit <http://www.titleix.utexas.edu/>, contact the Title IX Office via email at titleix@austin.utexas.edu, or call 512-471-0419.

Course Calendar:

Class	Dates	Topics*	Reading	Assignments Due Before Class
1	Jan 12 (M)	Introduction to Class		
2	Jan 14 (W)	Tour of the Universe	1.1-1.9	Module #1 Pre-class Homework
3	Jan 21 (W)	The Celestial Sphere	2.1	Module #2 Pre-class Homework
—	Jan 26 (M)	Ice-pocalypse 🧊🧊🧊🧊🧊		
4	Jan 28 (W)	Motion of the Stars	4.1	Module #3 Pre-class Homework
5	Feb 2 (M)	Motion of the Sun	4.2	Assessment HW #1: Motion of Stars
6	Feb 4 (W)	Seasons	4.2	Module #4 Pre-class Homework
7	Feb 9 (M)	Moon Phases	4.5	Module #5 Pre-class Homework Assessment HW #2: Motion of the Sun
Exam #1 on Tuesday Feb 10 @ 7-8p in WEL 3.502				
8	Feb 11 (W)	Predicting Moon Phases	4.5	
9	Feb 16 (M)	Models of the Universe	2.2, 4.7	Module #6 Pre-class Homework Assessment HW #3: Moon Phases
10	Feb 18 (W)	Copernicus through Newton	2.4, 3.1, 3.4, 3.2, 3.3	Module #7 Pre-class Homework
11	Feb 23 (M)	Process of Science & Nature of Light	5.1, 5.2, 5.3	Module #8 Pre-class Homework Assessment HW #4: Models of the Universe
12	Feb 25 (W)	Thermal Radiation, Atoms and Spectra	5.4, 5.5	Module #9 Pre-class Homework
13	Mar 2 (M)	Solar System	7.1, 7.2	Module #10 Pre-class Homework Assessment HW #5: Light and Atoms
19	Mar 4 (W)	Telescopes	6.1-6.6	
Exam #2 on Thursday Mar 5 @ 7-8 in WEL 3.502				
14	Mar 9 (M)	Climate Change	8.3, 8.4	Module #11 Pre-class Homework
15	Mar 11 (W)	Climate Change	8.3, 8.4	Module #12 Pre-class Homework Astronomy in the News (due 3/11)
Spring Break!!!				
16	Mar 23 (M)	Stars	16.1-16.3, 17.1-17.3, 18.4	Module #13 Pre-class Homework Assessment HW #6: Climate Change

Class	Dates	Topics*	Reading	Assignments Due <u>Before Class</u>
17	Mar 25 (W)	Stellar Evolution	22.1-22.5, 23.1-23.3	Module #14 Pre-class Homework
20	Mar 30 (M)	Exoplanets #1	5.6, 21.3, 21.4	Module #16 Pre-class Homework Assessment HW #7: Stars
21	Apr 1 (W)	Exoplanets #2	5.6, 21.3, 21.4	Module #17 Pre-class Homework Moon Journal (due 4/1)
Exam #3 on Thursday Apr 2 @ 7-8p in WEL 3.502				
22	Apr 6 (M)	The Milky Way and Nearby Galaxies	25.1-25.5, 26.1-26.3	Module #18 Pre-class Homework Assessment HW #8: Exoplanets
23	Apr 8 (W)	The Milky Way and Nearby Galaxies	25.1-25.5, 26.1-26.3	Module #19 Pre-class Homework
24	Apr 13 (M)	Hubble's Law	26.4, 26.5	Module #20 Pre-class Homework Assessment HW #9: Milky Way/Galaxies
25	Apr 15 (W)	Hubble's Law	26.5	Citizen Science Project (due Apr 15)
26	Apr 20 (M)	Distant Galaxies	28.1-28.3	Module #21 Pre-class Homework Assessment HW #10: Hubble's Law
27	Apr 22 (W)	Mysteries of the Universe	—	Assessment HW #11: Galaxy Evolution
28	Apr 27 (M)	The Beginning and End of the Universe	29.1-29.7	Optional Extra Credit: Observing, Module #22 Pre-class Homework
Exam #4 on Monday Apr 27th @ 7-8p in WEL 3.502				

****Schedule of topics subject to change***